



THE

TIDAL

UNIVERSE

Dr Das Baskill
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The background of the slide is a photograph of a sunset over the ocean. The sky is a deep orange, and the sun is a bright orange disk on the horizon. In the distance, several wind turbines are visible against the horizon. The ocean has small, dark waves. The foreground shows a sandy beach with some dark pebbles.

THE

TIDAL UNIVERSE

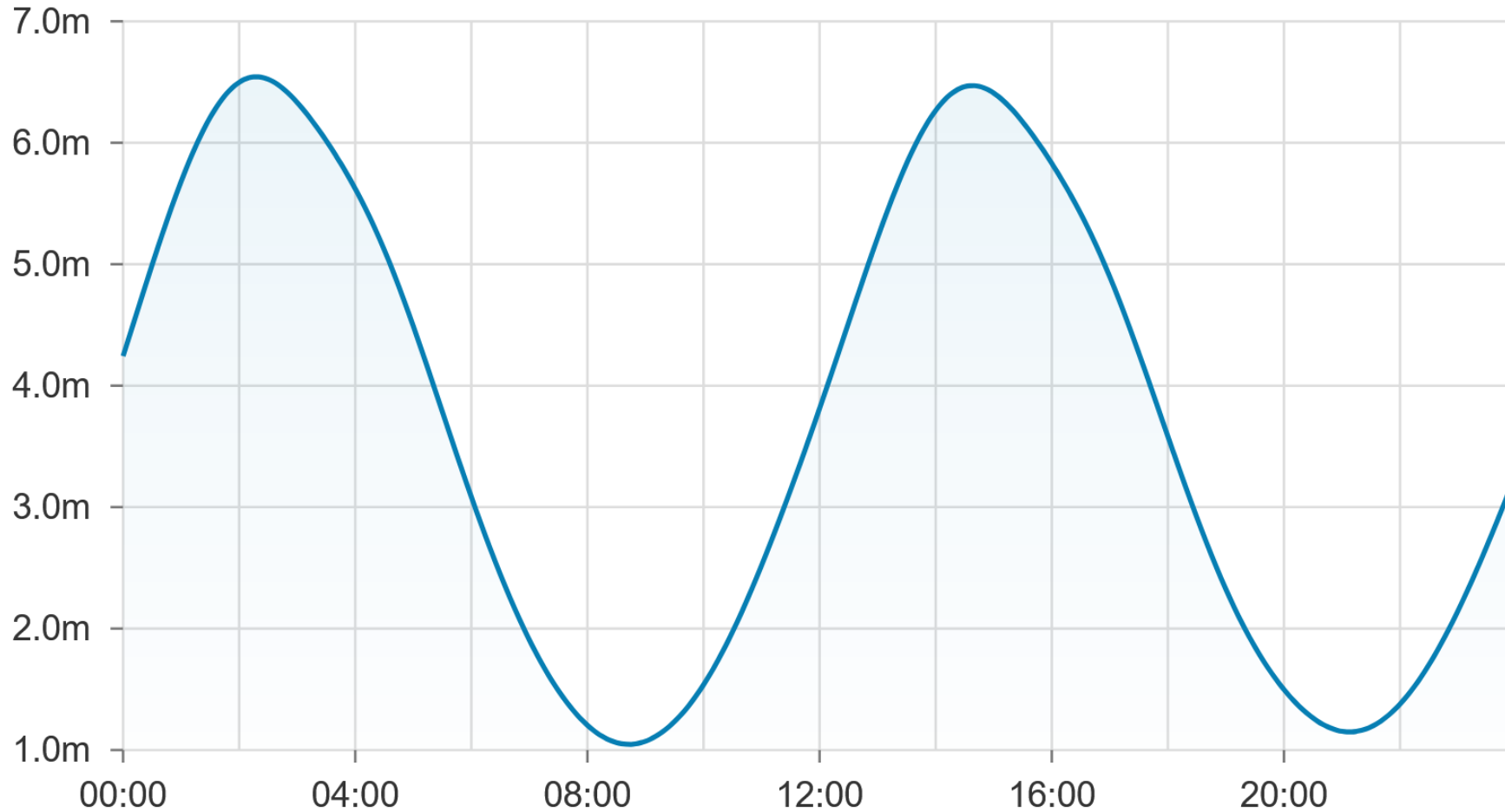
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Planet Earth is a water planet...

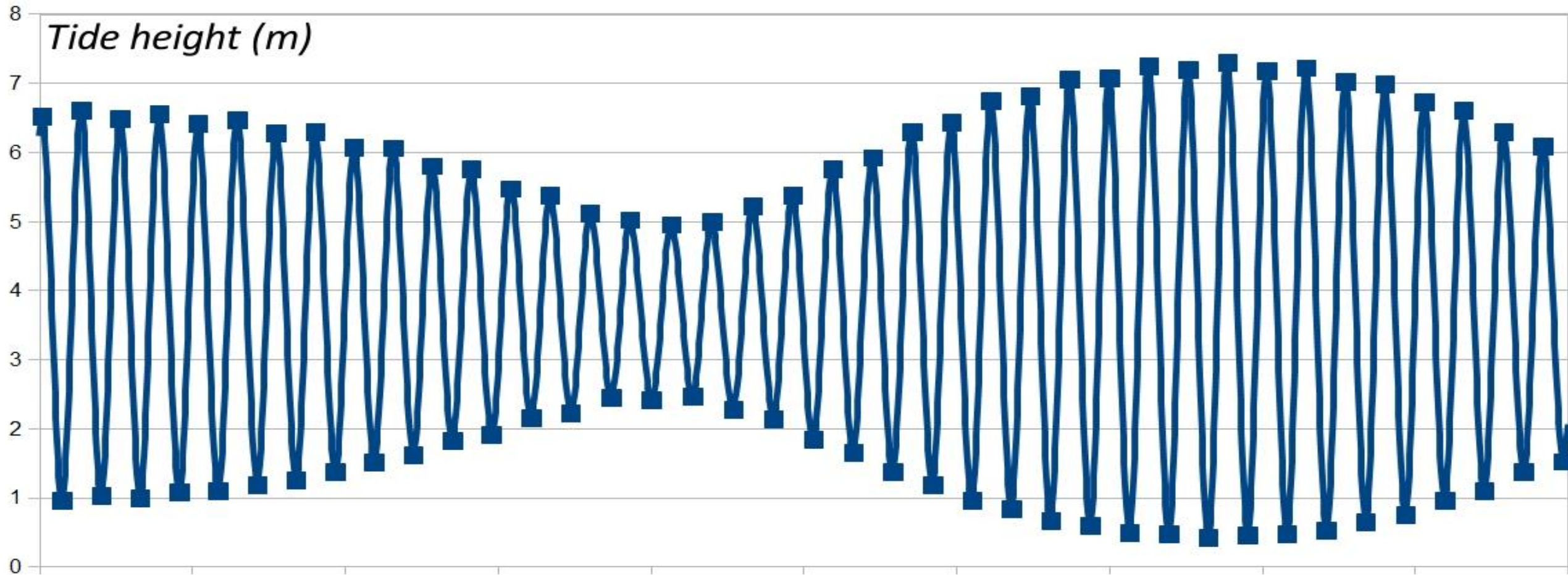


Defining tides



Tides are caused by the
gravitational interplay of the Sun ($\frac{1}{3}$) & Moon ($\frac{2}{3}$)

Defining tides



Extreme tides occur every fortnight. Variety due to the Sun & Moon's relative positions, complicated by the Moon's tilted orbit.



Scaled model of the Earth & Moon





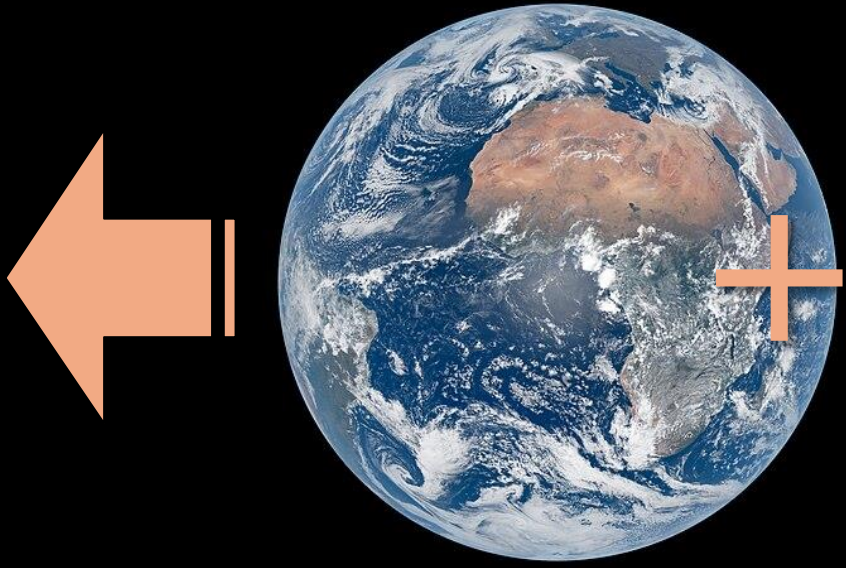
Newtonian gravity

A mathematical explanation of gravity.

$$F = \frac{G M_1 M_2}{r^2}$$

The factor $G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$

Centrifugal forces



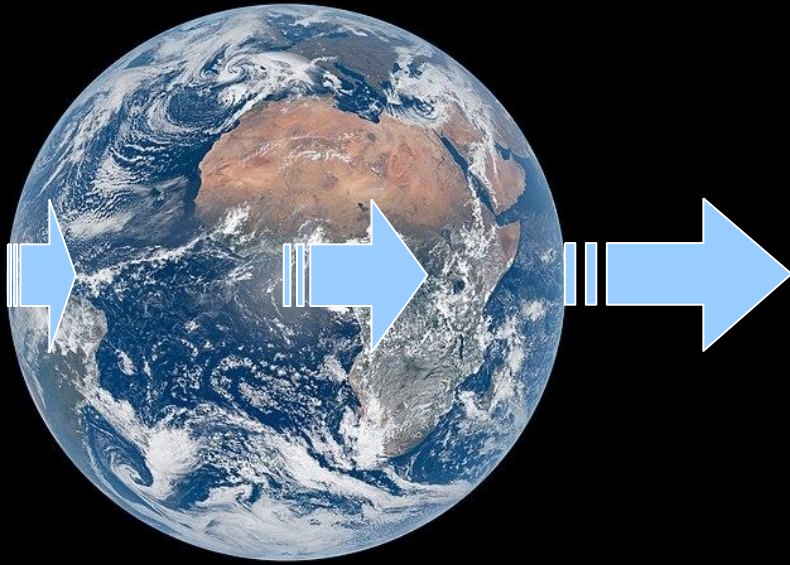
Earth



Moon

Not to Scale!

Gravitational forces



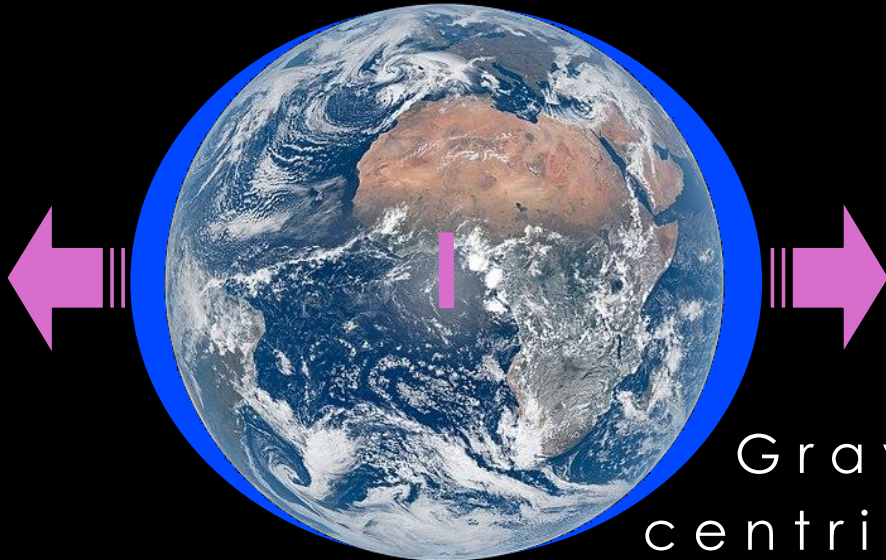
Earth

$$F_G = \frac{G M_E M_M}{r^2}$$

Gravity from
the Moon



Moon



Gravity and
centrifugal forces combined



The Earth rotates every 24 hours
beneath the Moon,
Resulting in **TWO** tidal bulges

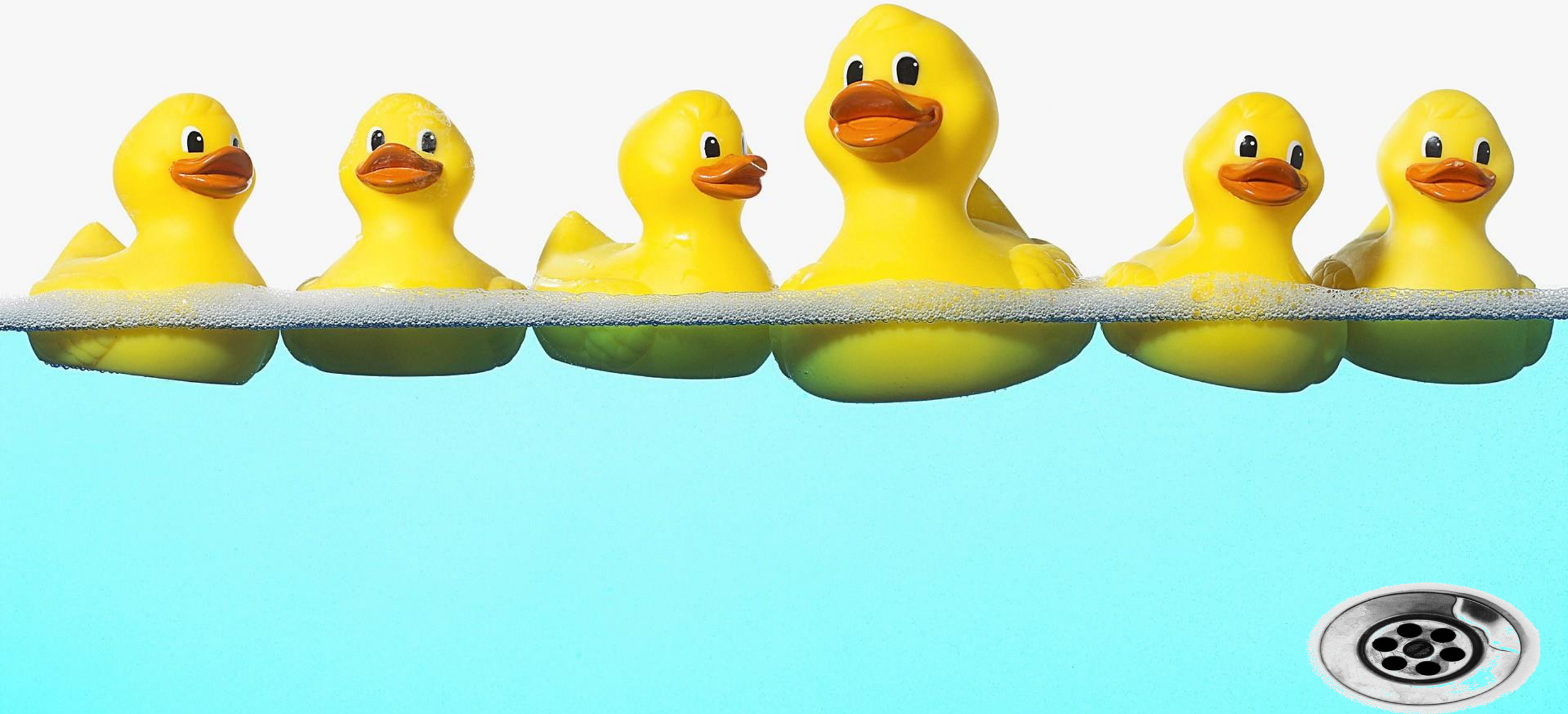
Earth

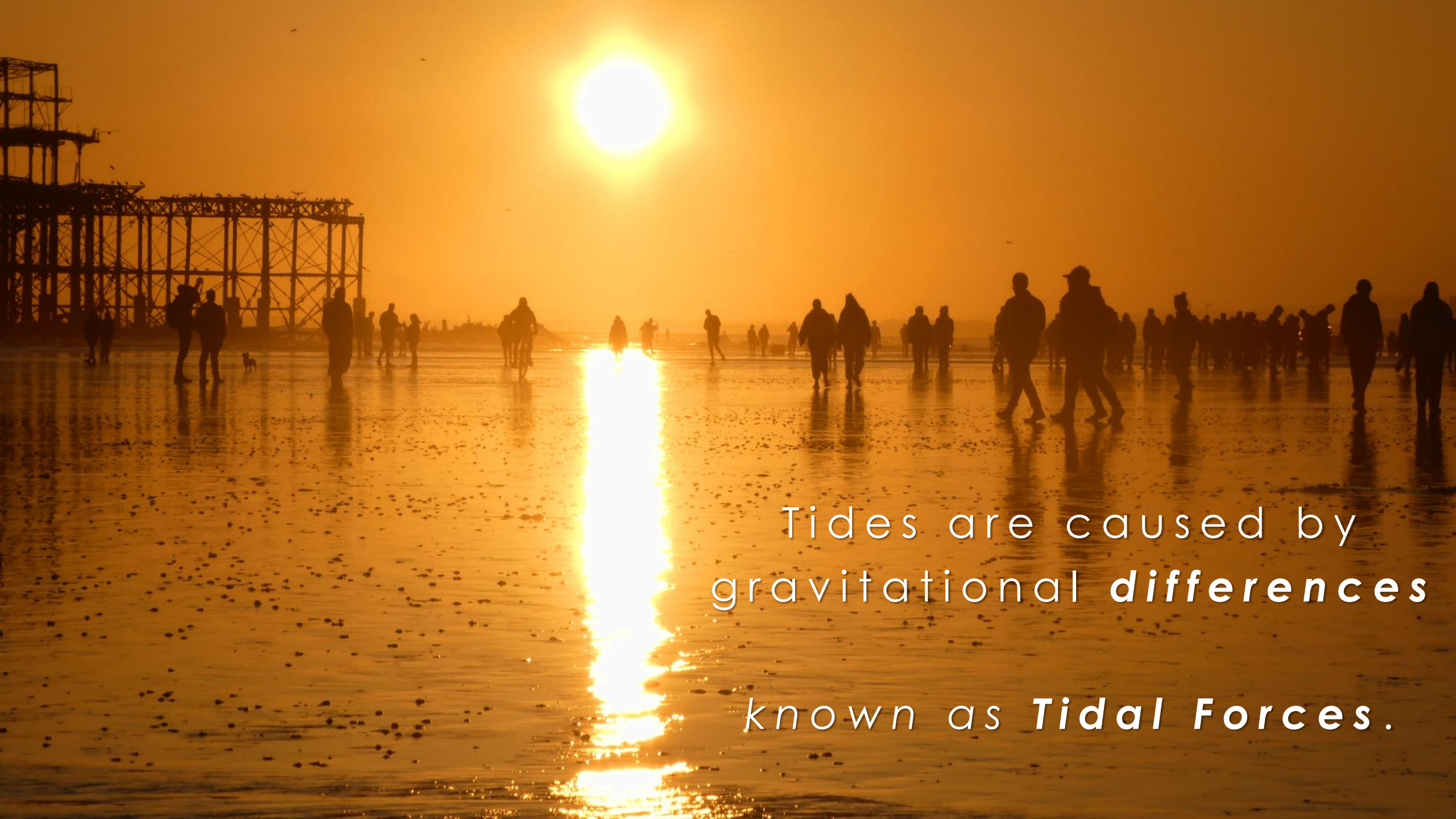


Moon



Analogy: Ducks in the bath!





Tides are caused by
gravitational ***differences***

known as ***Tidal Forces.***

The Moon also
causes the
Earth's surface
to rise & fall...
by 38cm!



Our Moon is
tidally locked



Tidal bulges

do not only occur

- ~~The Moon's~~ permanent bulge is 10-20m
- The Moon's surface also flexes by 10-20cm

...but across

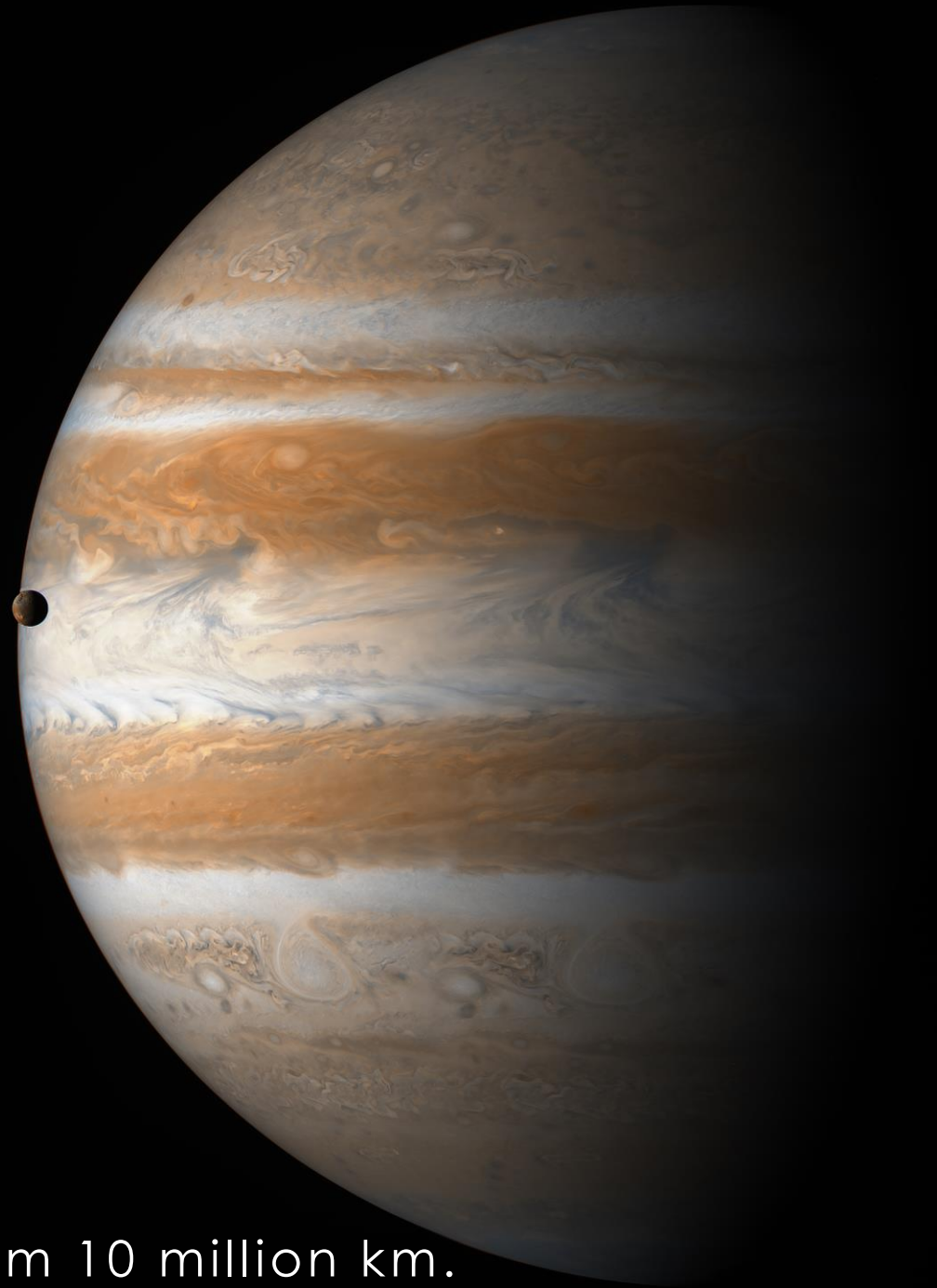
the entire Universe.



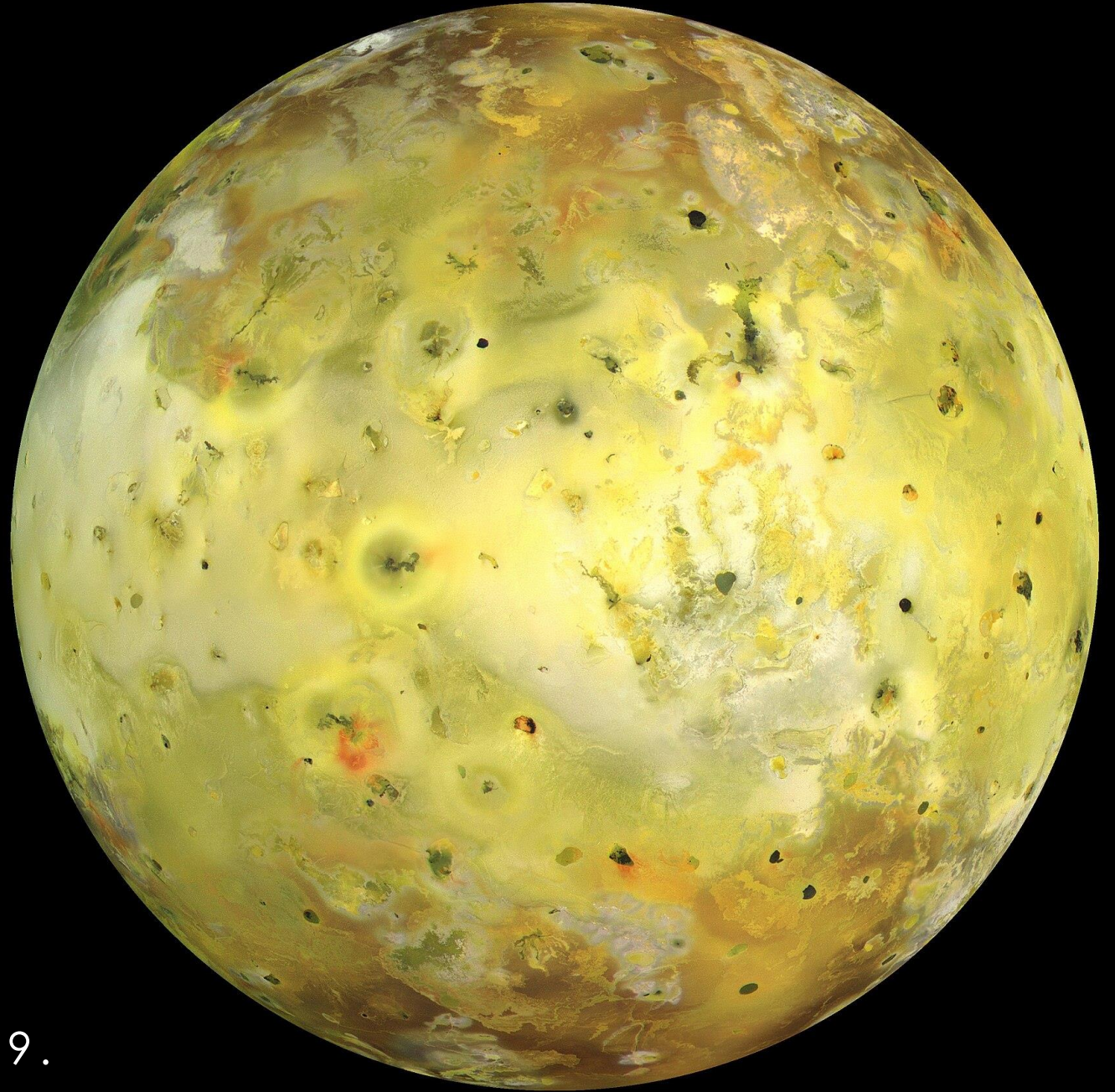
Tidal Moons

Io and Jupiter

observed by Cassini in 2001 from 10 million km.

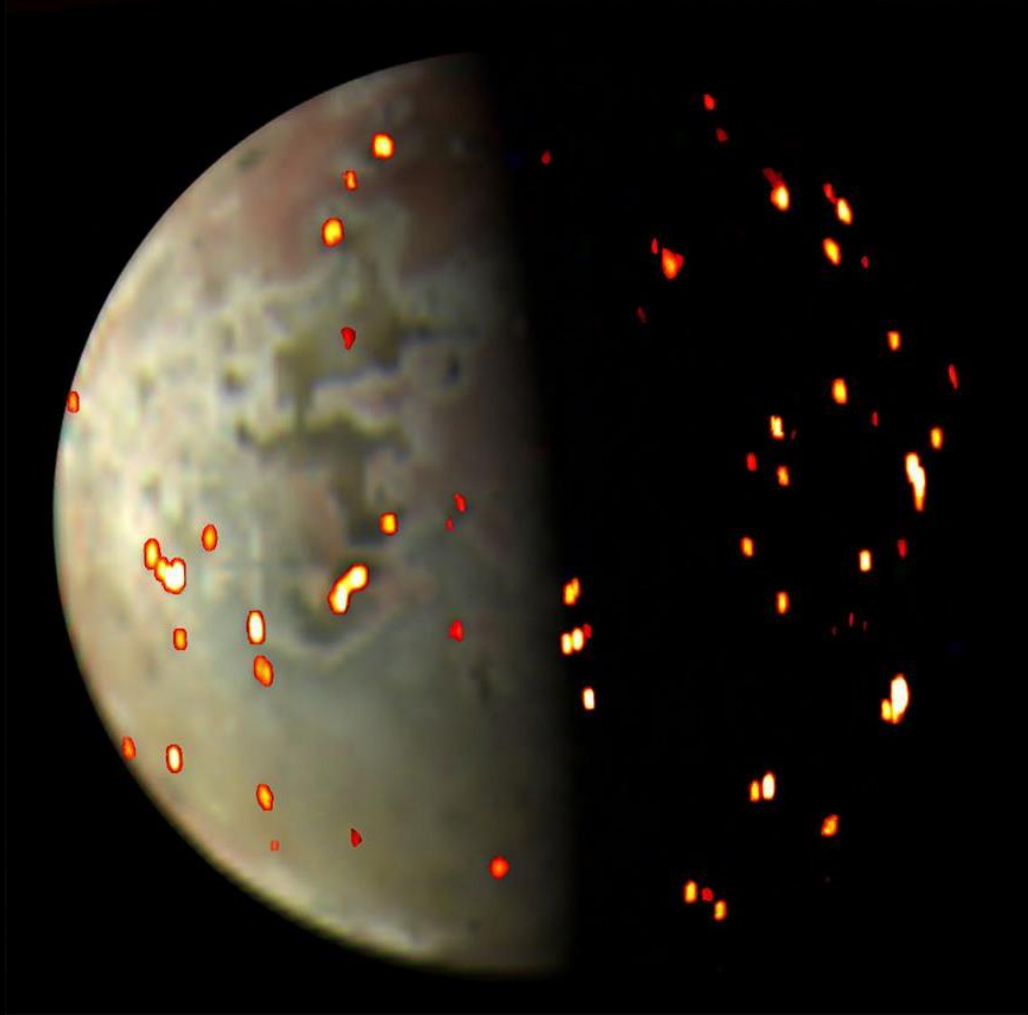


Tidal Moons: Io

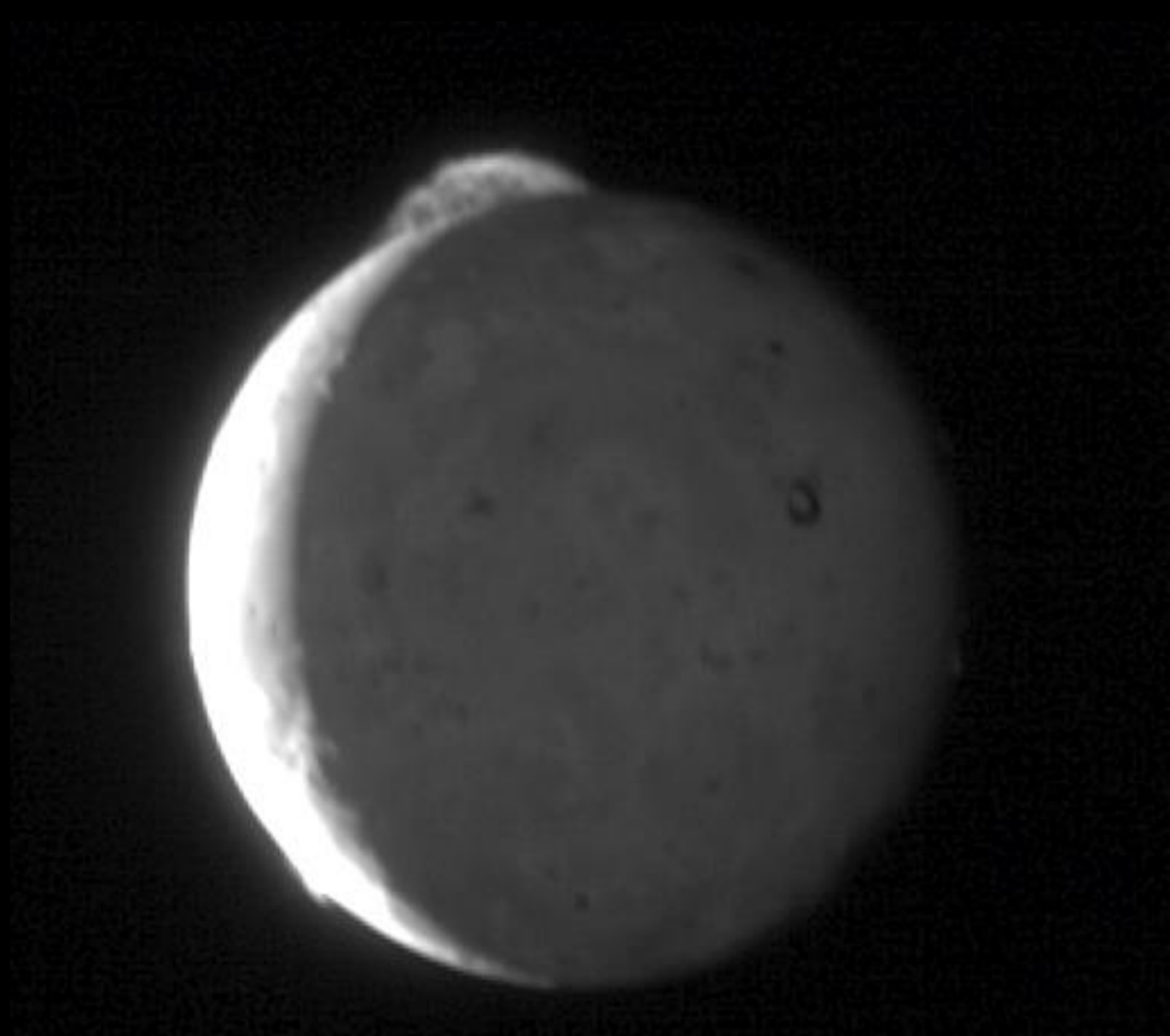


Io observed by Galileo in 1999.

Tidal Moons: Io

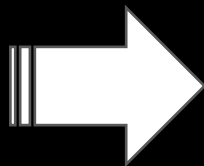


Volcanic activity on Io.
Juno spacecraft, 2023.



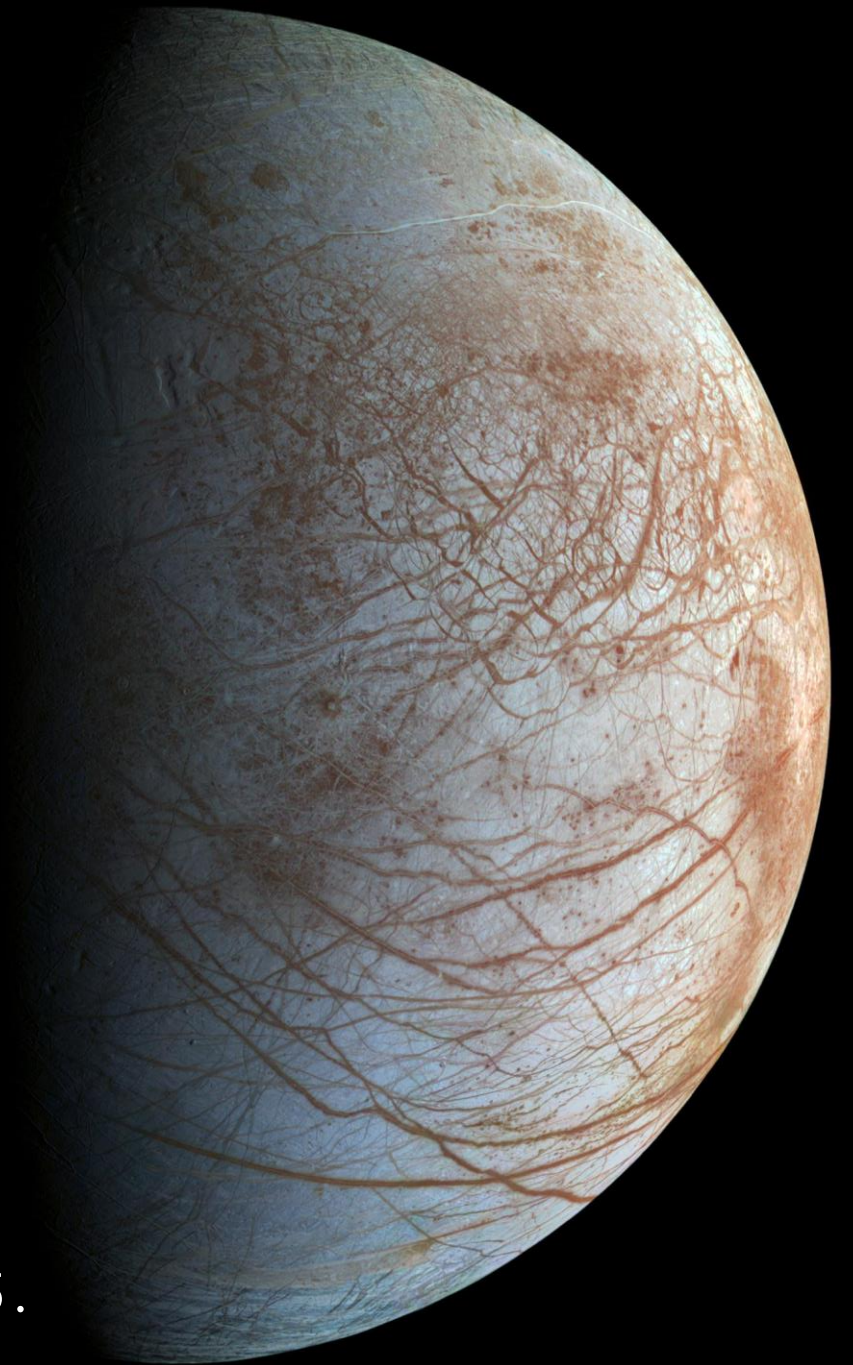
Five images over 8 minutes.
New Horizons, 2007.

Tidal Moons: Europa

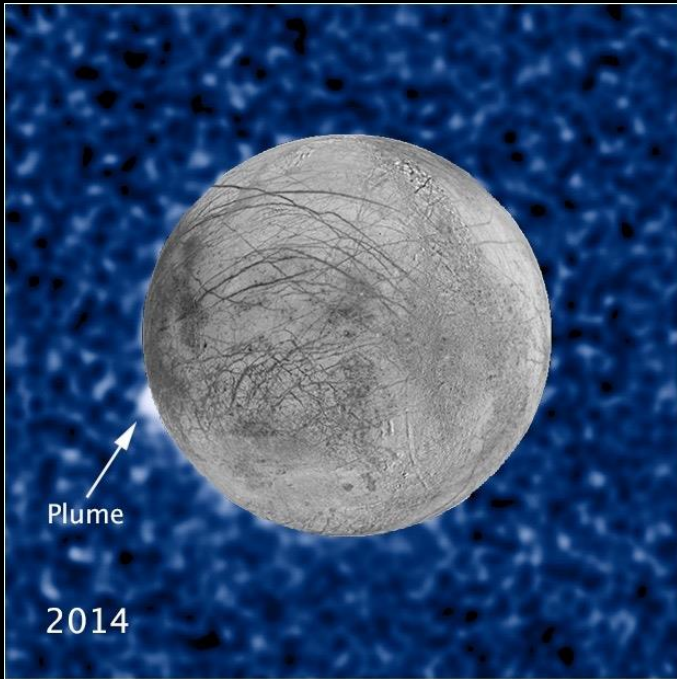


Tidal Moons: Europa

Jupiter's moon Europa,
observed by Galileo in 1995.



Tidal Moons: Europa

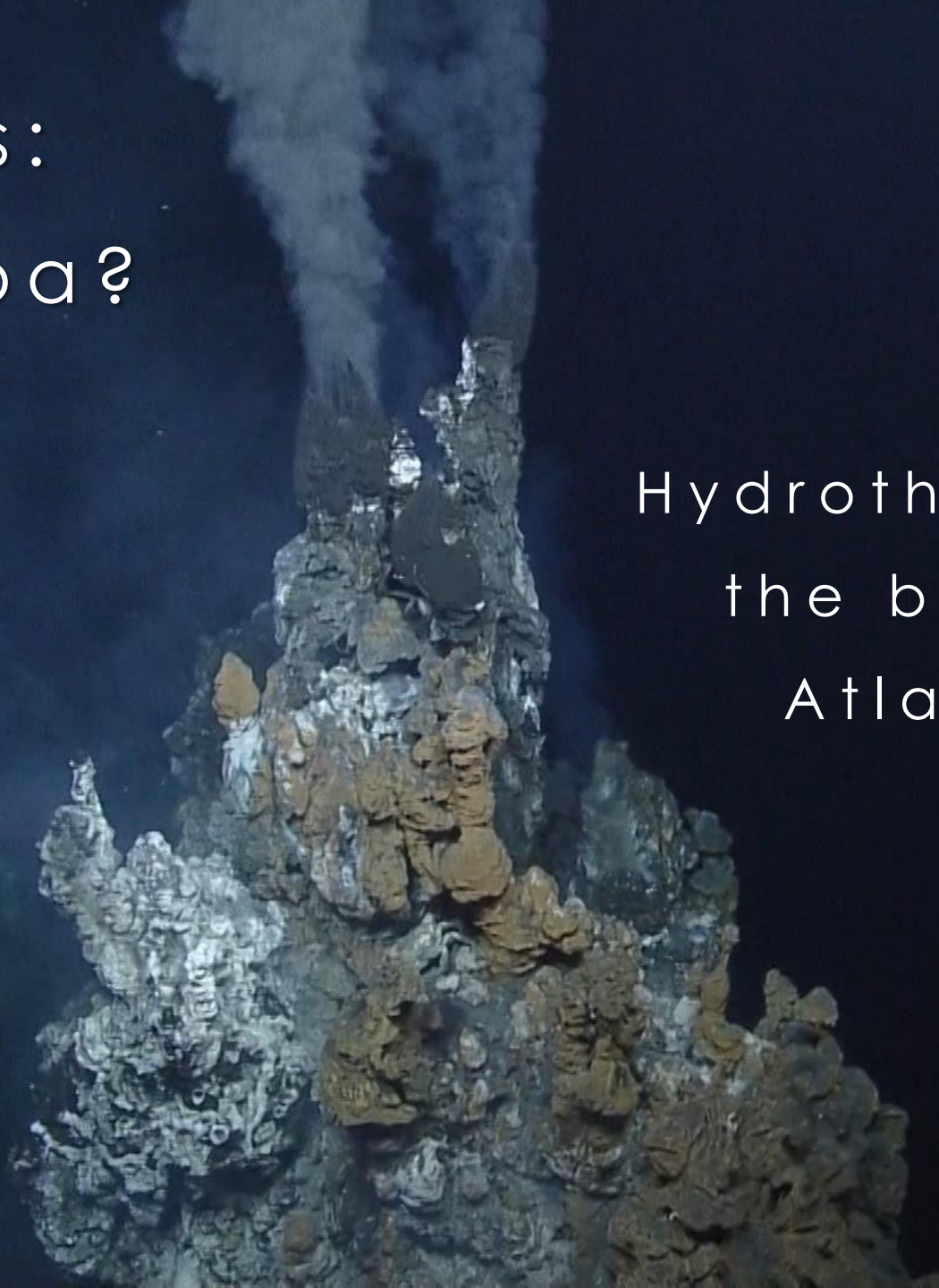


Jupiter's moon Europa,
observed by the
Hubble Space Telescope
in 2014 & 2016.



Tidal Moons: Europa?

Hydrothermal vents at
the bottom of the
Atlantic ocean



Tidal Moons:
Europa?

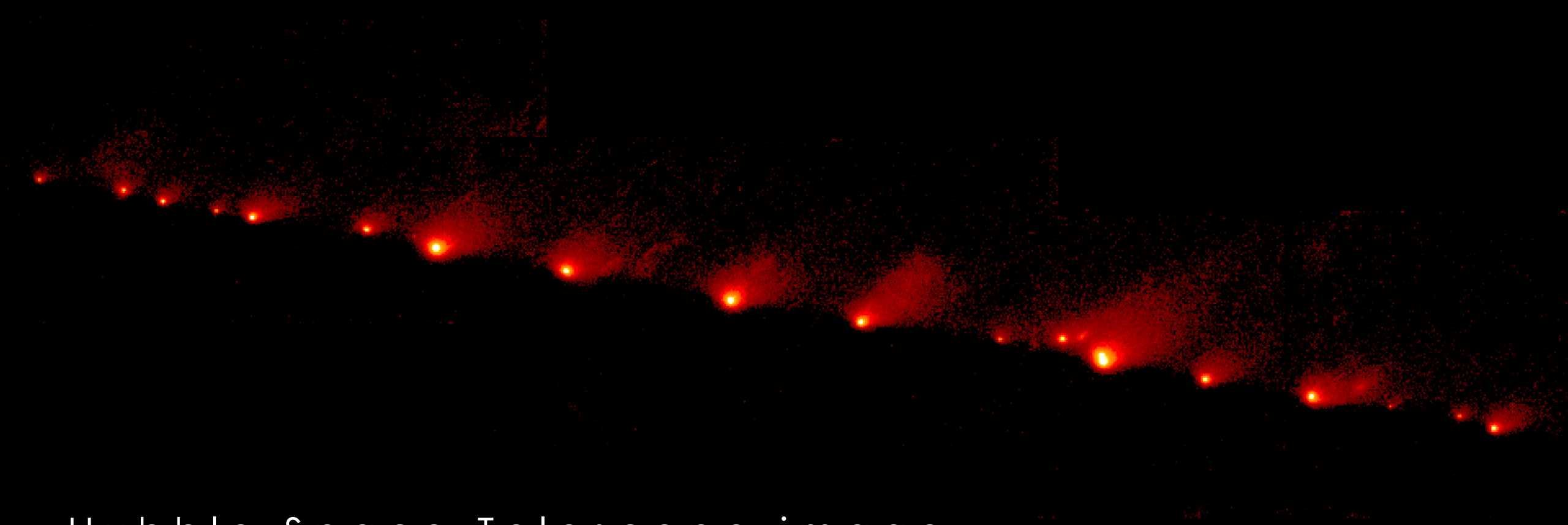


Tidal Comets

Comet C/2025 A6 (Lemmon),
October 2025.

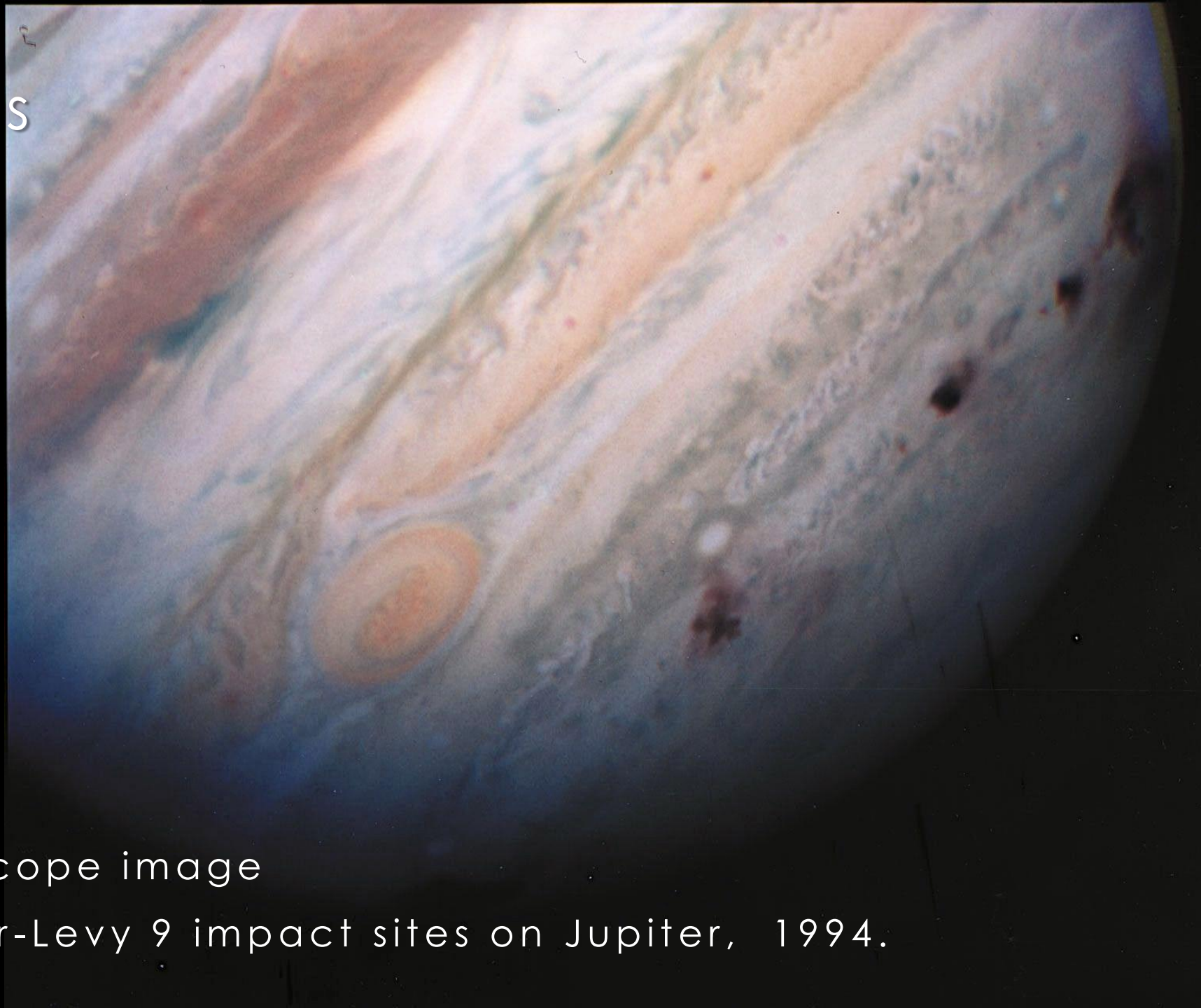


Tidal Comets



Hubble Space Telescope image
of comet Shoemaker-Levy 9, May 1994.

Tidal Comets



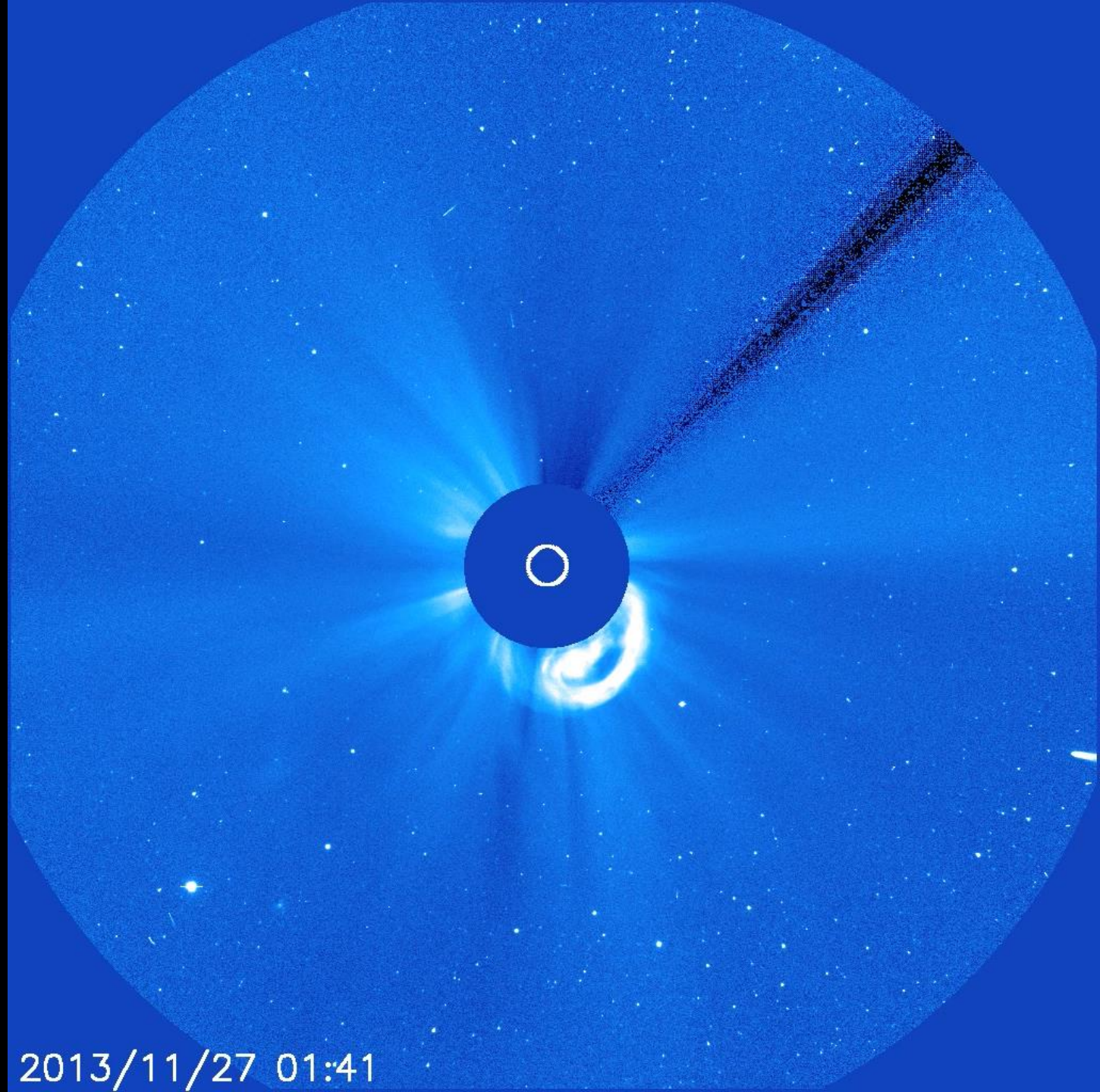
Hubble Space Telescope image
of comet Shoemaker-Levy 9 impact sites on Jupiter, 1994.

Tidal / Roche Limit

The Roche Limit is the distance where an object will disintegrate due to tidal forces.

e.g. Comet ISON, November 2013

Credit: SOHO observatory



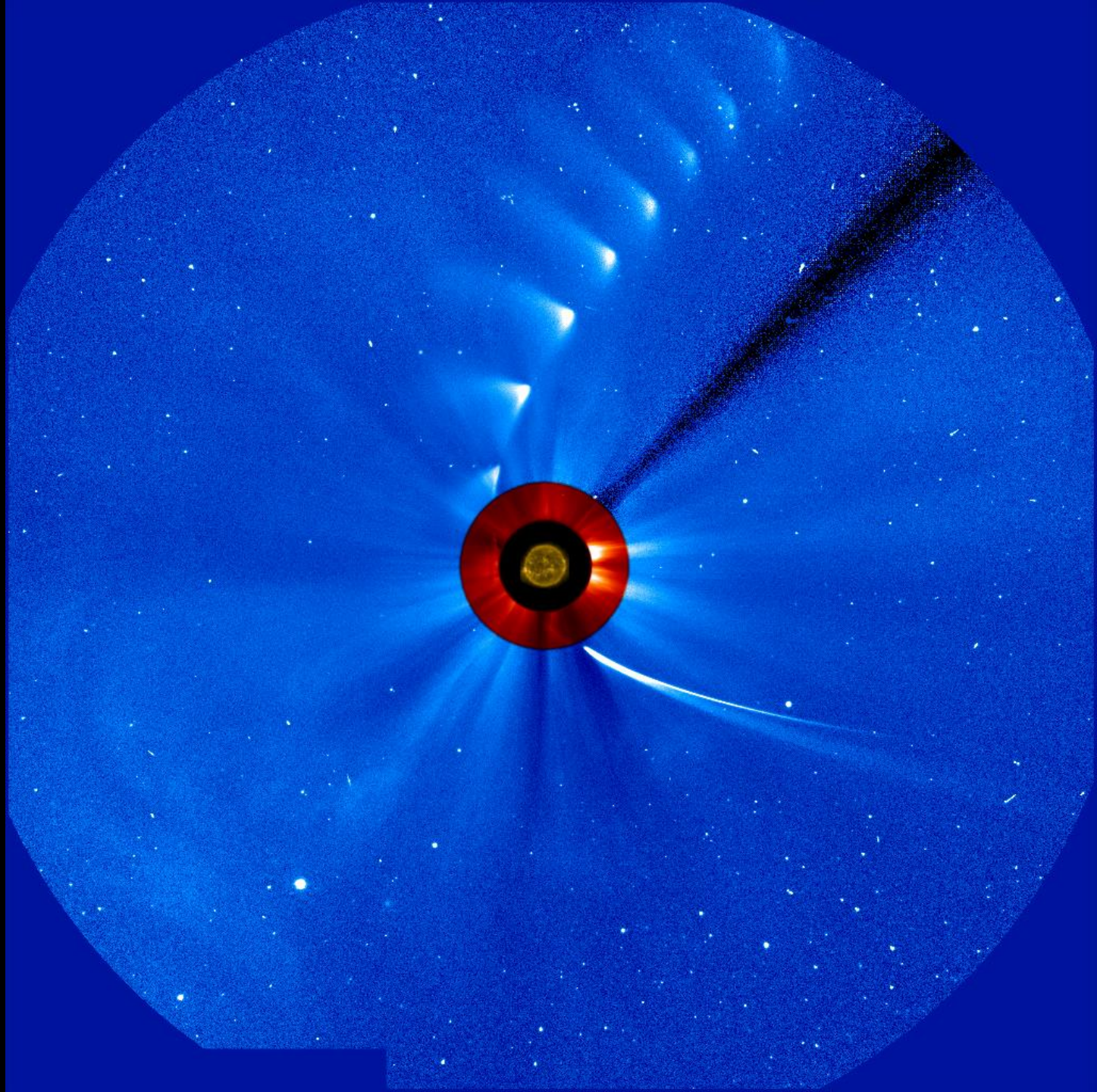
2013/11/27 01:41

Tidal / Roche Limit

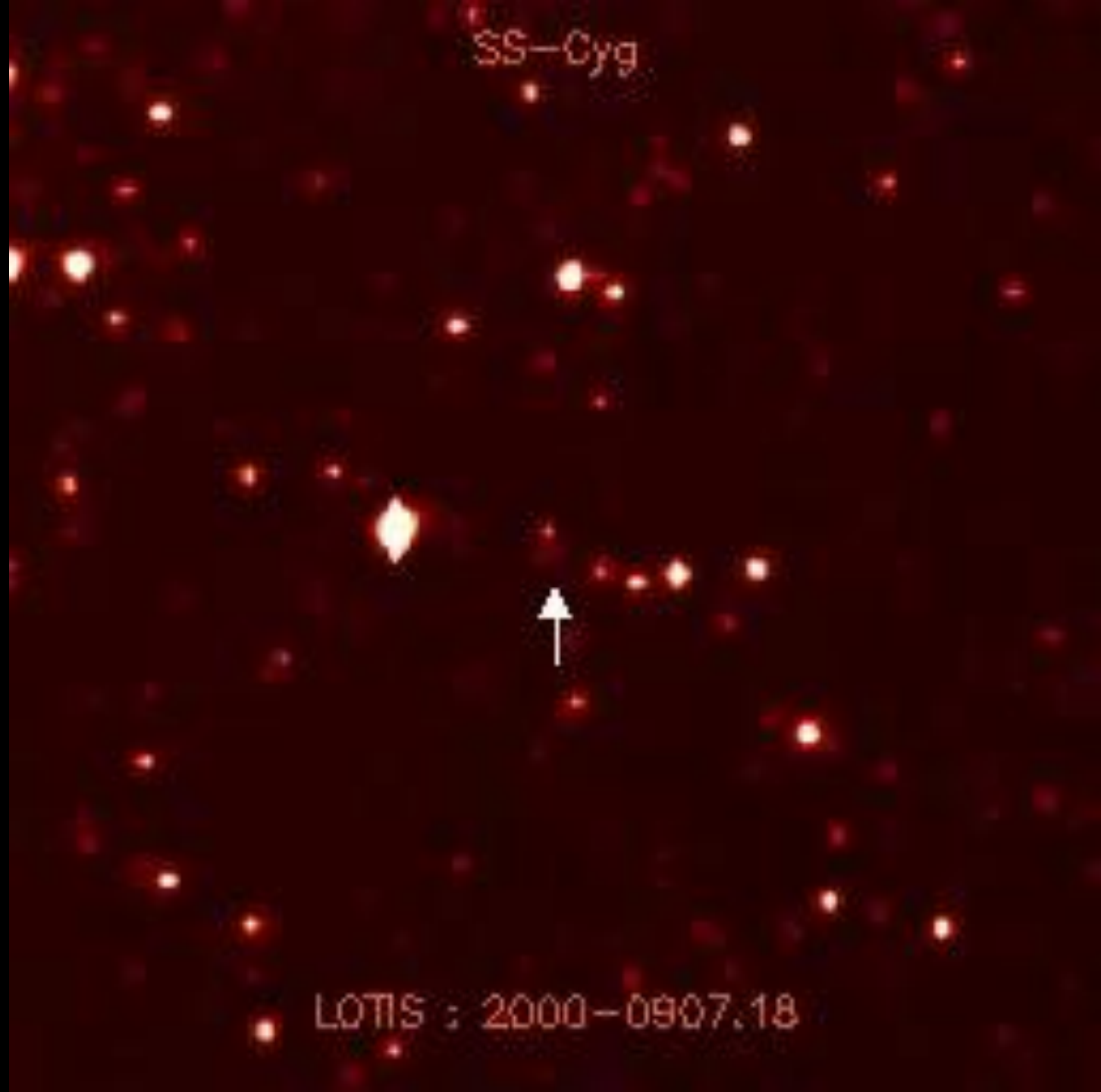
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Tidal Stars



Tidal Stars

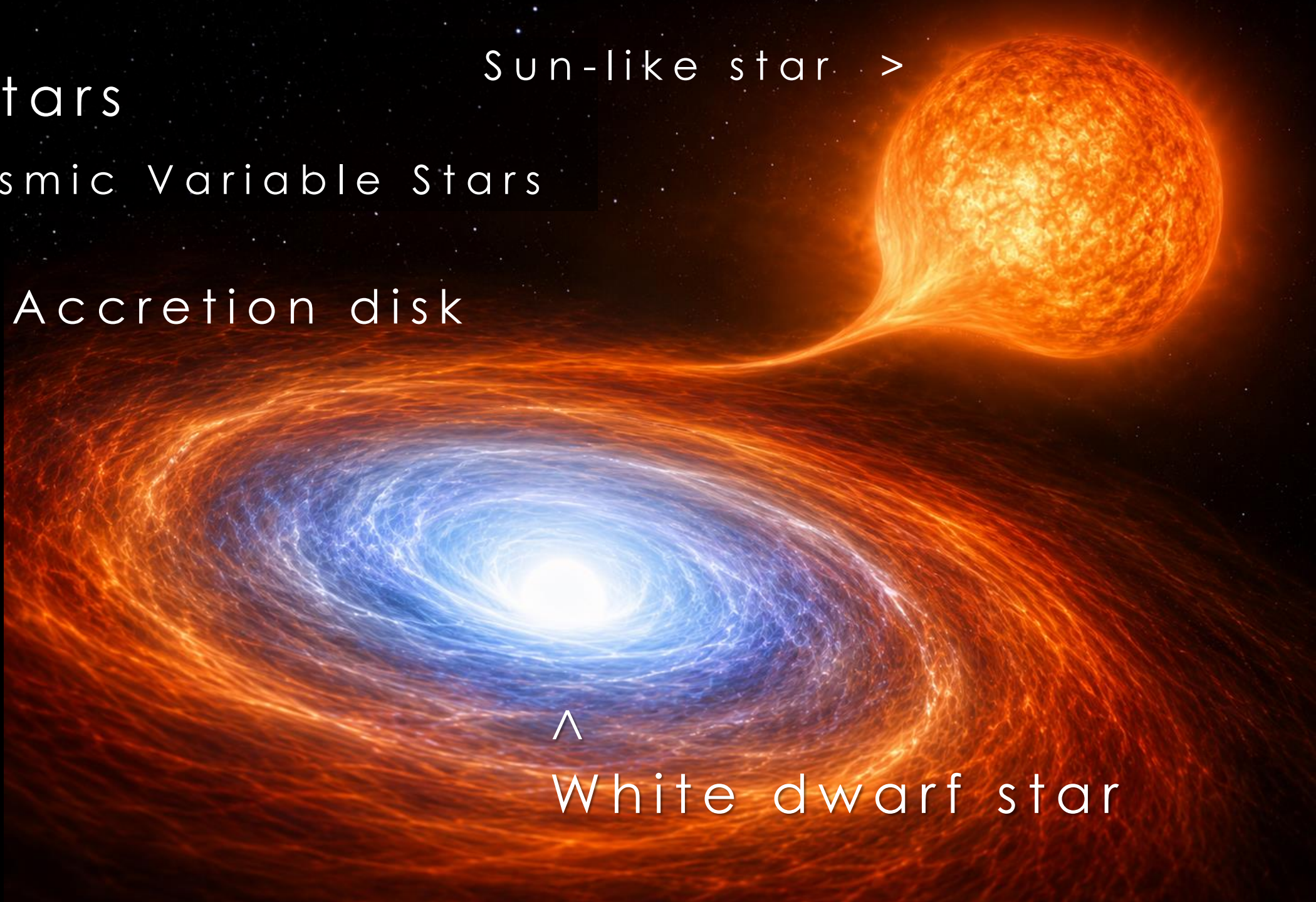
Cataclysmic Variable Stars

Sun-like star >

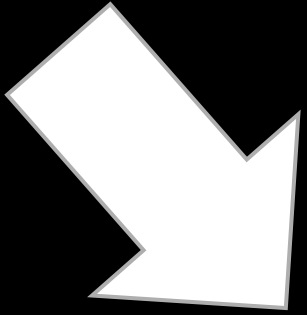
Accretion disk

^

White dwarf star



Tidal Stars: Black Holes



Artistic impression credit: Das Baskill (University of Sussex)

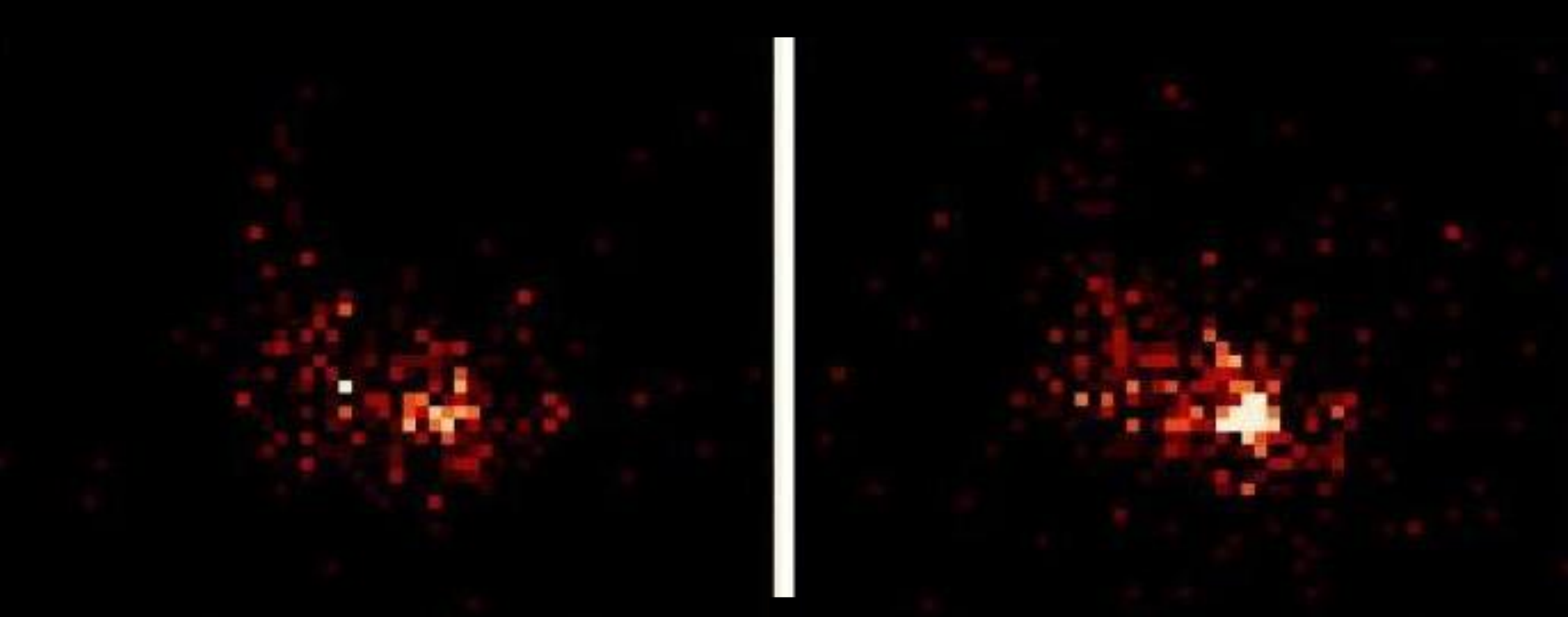
Impression of Cygnus X-1

Most probable nearby
black hole $\sim 6,000$ ly away,
or 60,000,000,000,000,000 km

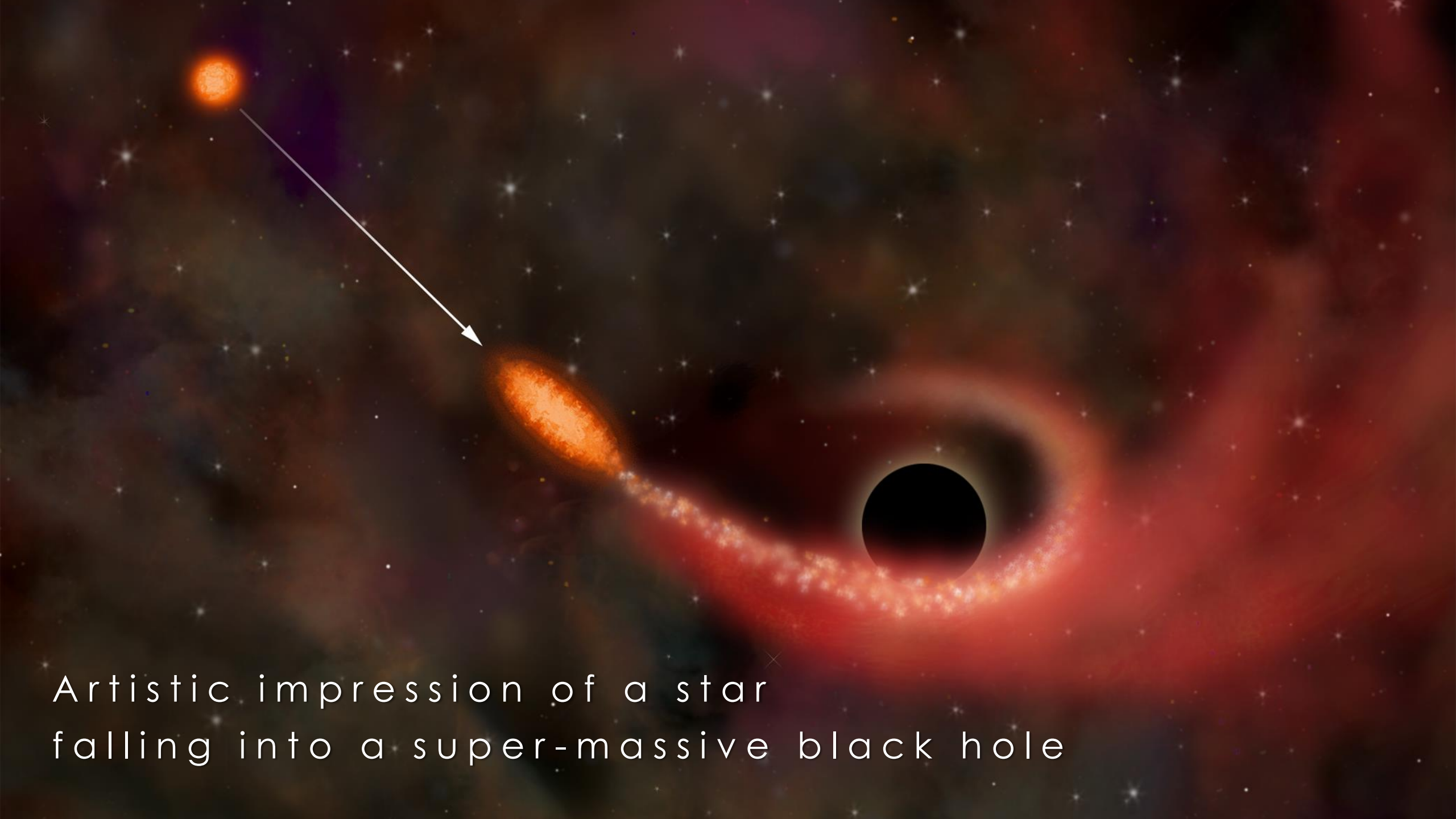
Credit: Martin Kornmesser (ESA/Hubble)



XMM-Newton



A star falling into Sagittarius A*
the black hole at the centre of our galaxy



Artistic impression of a star
falling into a super-massive black hole

Spaghettification

A 5 solar-mass BH would
spaghettify you at 3,000km.

A 4 Million solar-mass BH
would spaghettify
you at 300,000km

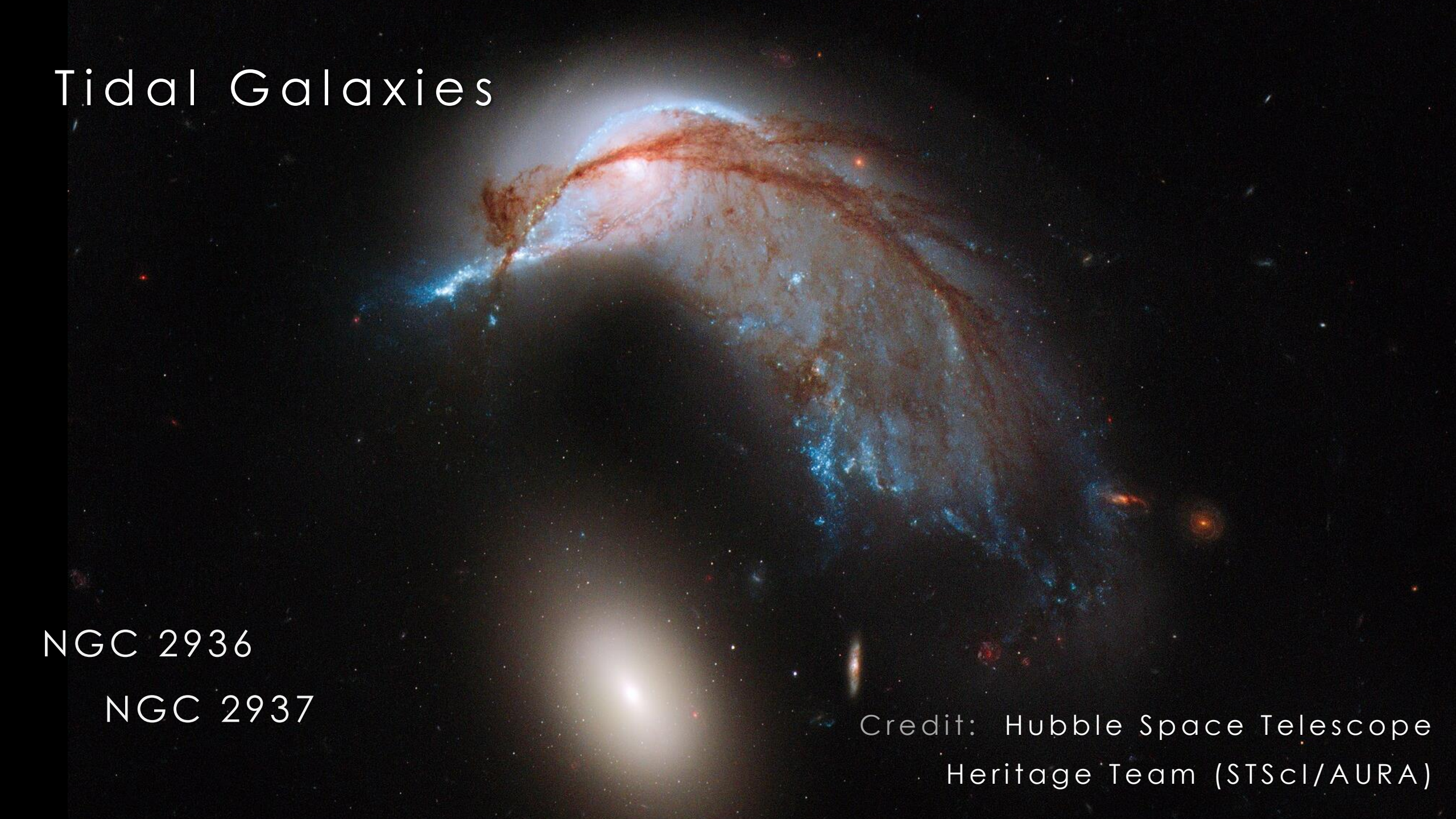


Tidal Galaxies

NGC 2936

NGC 2937

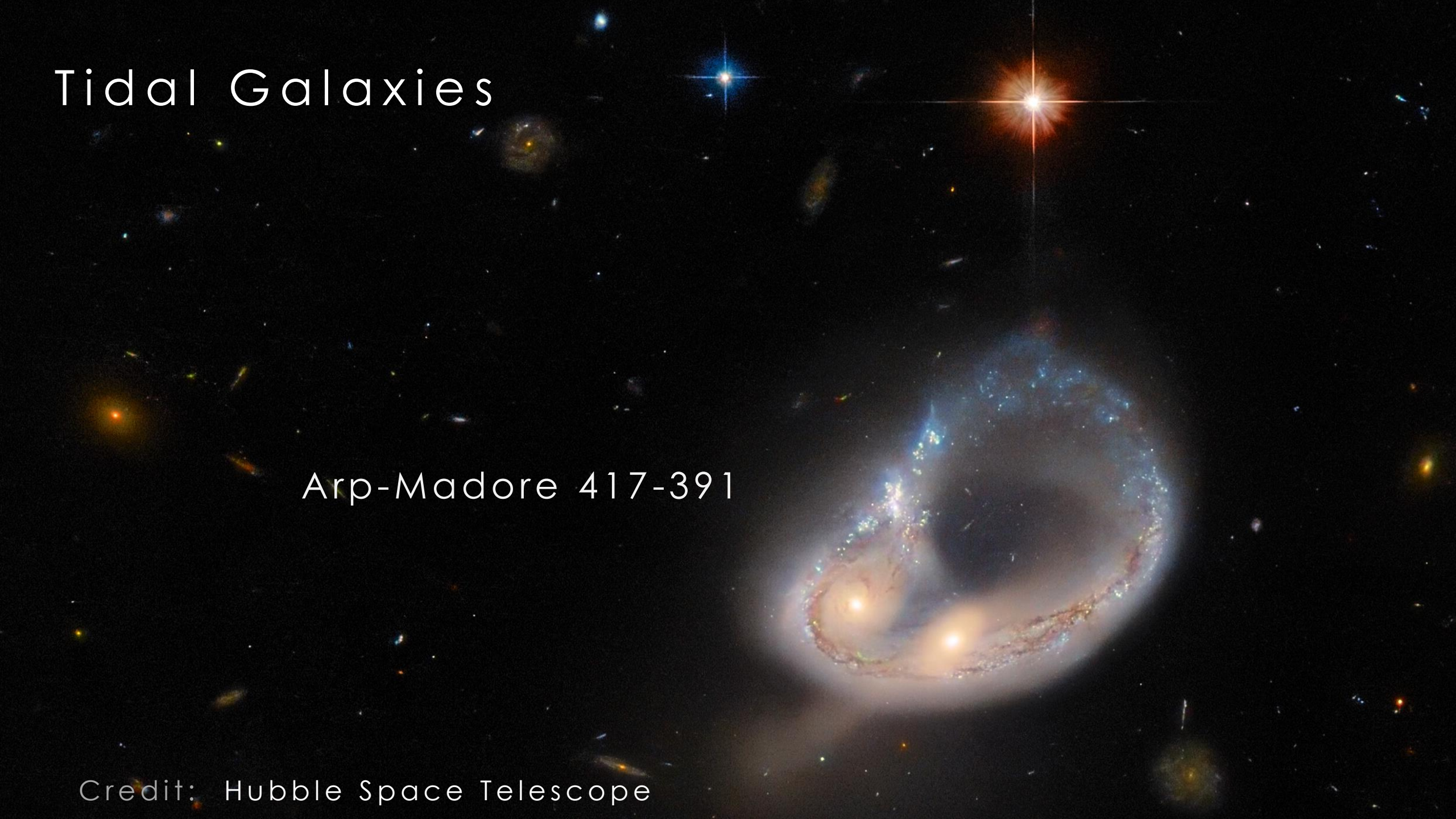
Credit: Hubble Space Telescope
Heritage Team (STScI/AURA)



Tidal Galaxies

Arp-Madore 417-391

Credit: Hubble Space Telescope



Tidal Galaxies

Credit: Subaru, NAOJ, NASA/ESA/Hubble and Colombari

Tidal Galaxies



Credit: Hubble Space Telescope

Tidal Galaxies

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Small Magellanic Cloud

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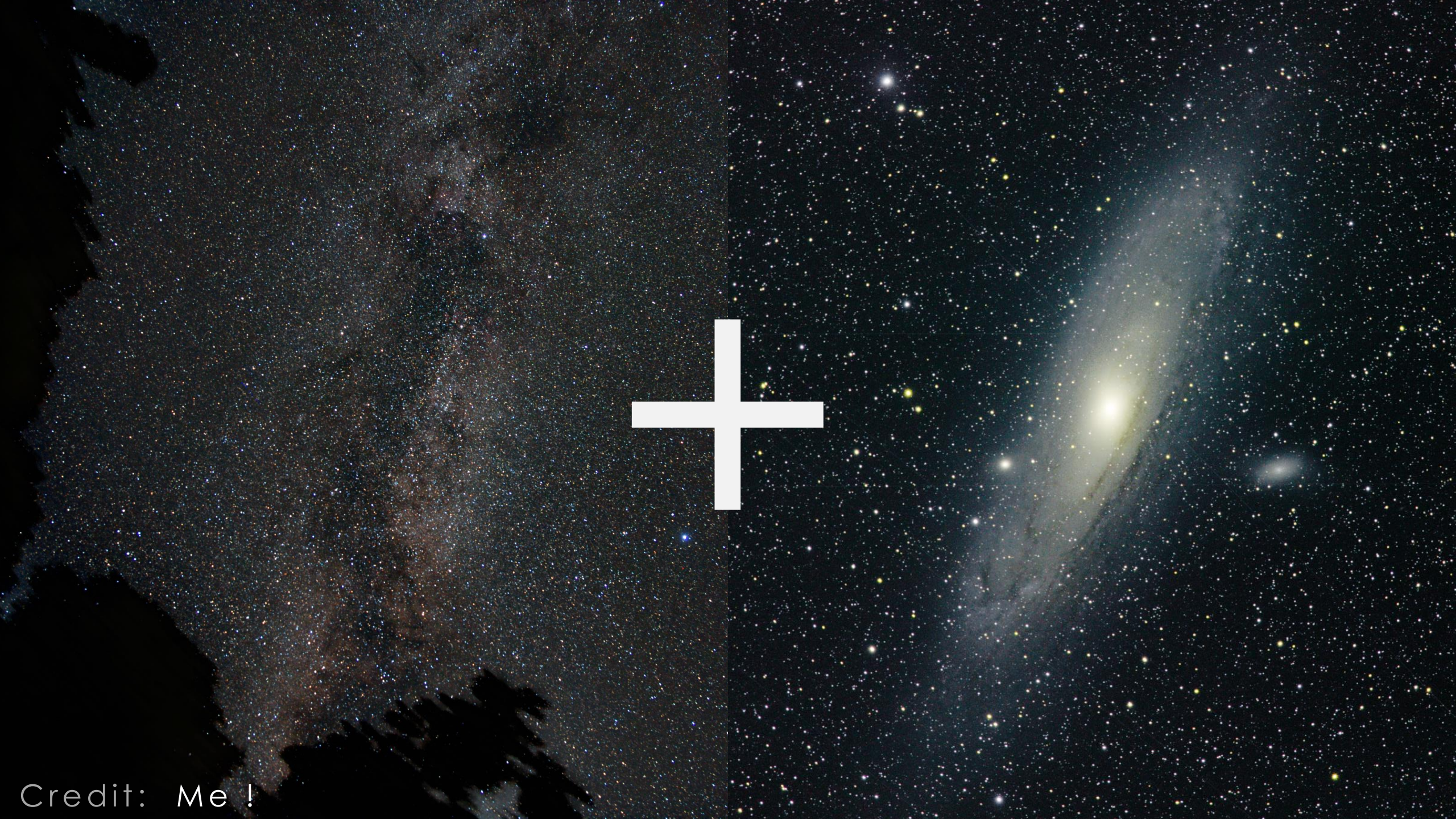
Large Magellanic Cloud

Credit: M Claro / ESO

Tidal Galaxies



Credit: ESO / Beletsky



Credit: Me !



0.000 billion years





A sunset over the ocean with the title 'THE TIDAL UNIVERSE' overlaid. The sun is a bright yellow orb in the upper right, casting a long, shimmering reflection across the dark water. The horizon is visible in the distance, and the foreground shows dark, silty water with some small rocks or debris. The overall color palette is dominated by warm oranges, yellows, and dark blues.

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